

GNN-BERT Music Context Understanding — report outline

> **Status: skeleton.** Every number below is a placeholder marked `[TBD]` and
> must be filled from `results/metrics.json` after the real training runs.
> Nothing here is copied from the assignment PDF's Table 3 — those figures are
> instructor-written illustrations, not results.
>
> Rebuild the PDF with `python report/build\_report.py`, which renders this
> outline together with every figure currently in `results/plots/`.

1. Introduction

Music context understanding as a joint audio-structure and language problem.
What "context" means here: not just what instruments are present, but how a track is organised in time and how that organisation relates to how people describe it.

Contributions

1. A segment-graph representation of musical structure with a frozen 96-dim node feature contract.
2. Four tasks over one shared representation: text tagging, graph tagging, cross-attention fusion with multi-task emotion regression, and contrastive audio-graph $\leftrightarrow$ caption retrieval.
3. A fair-comparison protocol: parameter-matched CNN baseline, artist-disjoint splits, validation-only thresholds, and a degree-preserving rewiring control.

2. Related work

Music auto-tagging (MTAT lineage) · graph neural networks for music structure · text-audio contrastive retrieval · multi-task emotion regression (DEAM).

3. Data

Dataset	Nominal	Usable	Survival	Role
MagnaTagATune	25,863 clips	21,361 with ≥ 1 top-50 tag	82.6%	T1, T2, T3 tags
FMA-small	8,000	8,000 100%	genre probe, t-SNE colouring	
DEAM	1,802	1,802 100%	valence/arousal (T3)	
MusicCaps	5,521 rows	2,781	50.4%	captions (T4)
Lakh MIDI Clean	~17,256	17,184	99.6%	chord-estimator validation

MusicCaps attrition is the headline data fact. Roughly half the YouTube sources are gone. Per-split survival: **1,300/2,663 train (48.8%)** and **1,481/2,858 eval (51.8%)**. The eval survivor count *is* the Task 4 retrieval gallery size, so every R@K in Section 6 must be read against **N = 1,481**. Method: the source CSV is never modified; a manifest is derived by inner-joining against files that exist and decode, with a per-ytid status log in `data/splits/musiccaps\_download\_log.csv`.

Split integrity. MagnaTagATune's canonical hex-folder split (dirs `0-b` train, `c` val, `d-f` test) turns out **not** to be artist-disjoint: 57 artists

have clips in more than one folder. Repairing this by moving each such artist wholesale into its majority split relocates **3,465** of 21,361 clips (16.2%). We report results on the repaired, artist-disjoint split and note that numbers from the unrepaired split are optimistic by an unknown margin.

Chord estimation validation. Template matching on chroma agrees with Lakh MIDI symbolic ground truth on **[TBD]**% of frames (root-only: **[TBD]**%) over **[TBD]** files — see `results/chord_validation.json`.

4. Method

4.1 Segment graphs

Nodes are 3 s windows at 50% overlap (4–32 per track). The 96-dim node feature is fixed by contract: pooled mel mean/std (16+16), MFCC mean/std (20+20), chroma mean (12), spectral contrast mean (7), and five low-level descriptors.

Edges are bidirectional and of two kinds: a temporal chain, and k-NN similarity edges with `k = 4`. `edge_attr = [is_temporal, cosine_sim]`.

Why k-NN and not a cosine threshold τ : a single global τ produces isolated nodes on through-composed tracks and near-cliques on loop-based ones, so the topology encodes repetitiveness rather than segment relatedness and the receptive field varies across the batch.

4.2 Encoders

DistilBERT/BERT text encoder (CLS + token states); 2-layer GraphSAGE or GATv2 with mean||max readout. *Why only two layers:* on 4–32 node graphs, three or four rounds of message passing make every node's receptive field the whole graph and the states oversmooth.

4.3 Fusion (Task 3)

Six variants for the ablation: ``bert_only``, ``gnn_only``, ``early_concat``, ``late_concat``, ``cross_attention``, ``gated``, ``bidirectional``. The default treats the graph vector as a query token attending over caption tokens.

4.4 The masked multi-task loss

No track carries both MTAT tags and DEAM valence/arousal. BCE is computed only over cells where ``y_tags` != -1`; MSE only where the target is not ``nan``; each term is divided by an EMA of its own magnitude before weighting, because raw MSE on a 1–9 scale otherwise swamps per-tag BCE. Training alternates tag-bearing and emotion-bearing batches so both heads receive gradient inside every optimiser window.

4.5 Contrastive retrieval (Task 4)

Symmetric InfoNCE with a learnable, clamped log-temperature. Text embeddings are precomputed with a frozen encoder so a 512-sample batch fits on 4 GB.

5. Experimental setup

Hardware: GTX 1650 (4 GB, sm_75, no tensor cores), i5-11400H, 16 GB RAM.
AMP everywhere; batch 8×4 accumulation = effective 32. Three seeds (42, 1337, 2024); all results reported as mean $\pm$ sd.

****Fairness measures.**** B2's channel widths are searched so its parameter count matches the Task 2 GNN ([TBD] vs [TBD] parameters). Thresholds are tuned on validation, frozen, then applied once to test. Normalisation statistics come from the train split only. Splits are artist-disjoint and asserted at the top of every run.

6. Results

6.1 Tagging (macro-F1, micro-F1, AUC-PR — never accuracy)

Model	macro-F1	micro-F1	AUC-PR	params
B1 random (train prior)	[TBD]	[TBD]	[TBD]	0
B1 majority	[TBD]	[TBD]	[TBD]	0
B2 mel CNN (param-matched)	[TBD]	[TBD]	[TBD]	[TBD]
B3 / T1 BERT-only	[TBD]	[TBD]	[TBD]	[TBD]
B4 PCA + MLP	[TBD]	[TBD]	[TBD]	[TBD]
T2 GNN	[TBD]	[TBD]	[TBD]	[TBD]
T3 fusion	[TBD]	[TBD]	[TBD]	[TBD]

> Note for the discussion: the majority baseline reaches ~94.6% element accuracy
> on MTAT while scoring ~0.00 macro-F1. That contrast is why accuracy is absent
> from this table.

6.2 Emotion regression (DEAM)

Target	MAE	RMSE	R ²
valence	[TBD]	[TBD]	[TBD]
arousal	[TBD]	[TBD]	[TBD]

6.3 Retrieval (Task 4), gallery size N = 1,481

Direction	R@1	R@5	R@10	medR	MRR
graph → caption	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
caption → graph	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

6.4 Ablation

Seven fusion modes plus the ****rewired-graph control****. If macro-F1 survives degree-preserving rewiring, the topology carried no signal.

6.5 Representation analysis

t-SNE by genre and by mood quadrant, each with its k-NN probe and silhouette (the numbers, not the picture, are the evidence). Graph coherence $S_{\text{graph}} = (1/|E|) \sum 1[\cos(h_i, h_j) > \tau]$, trained vs rewired.

7. Qualitative analysis

Five BERT attention heatmaps; three Task 3 case studies (at least one a documented failure); ten retrieval examples (at least two failures).

8. Human evaluation

≥ 5 listeners rate caption-clip match on 1–5, with randomised presentation order and injected mismatched control pairs. Report Krippendorff α (ordinal), mean pairwise Spearman, and **control discrimination** — the gap between real and control ratings. Without that gap the ratings do not show raters were listening.

9. Limitations

MusicCaps attrition halves the gallery and biases it toward still-available videos · MTAT's 50-tag vocabulary is noisy and long-tailed even after synonym merging · chord estimation is template matching, not transcription · 4 GB of VRAM caps batch size and model scale · single-dataset emotion labels.

10. Conclusion

[TBD]

Reproducing every number here

```
```bash
make verify-data
make splits && make features
make all-tasks DEVICE=cuda
make baselines
make evaluate # writes results/metrics.json and results/plots/
python report/build_report.py
```
```

Appendix: machine-readable results

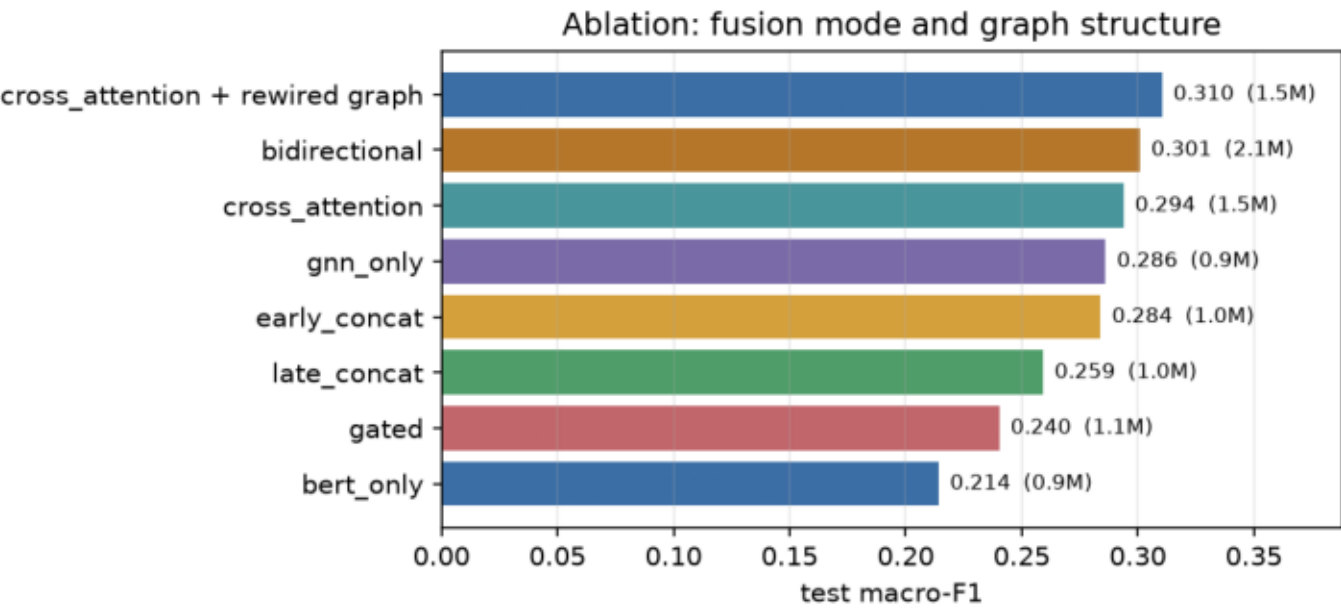
Source: `results/metrics.json` (synthetic data, seed 42, device cpu).

Tasks with saved runs: ['task1', 'task2', 'task3', 'task4']

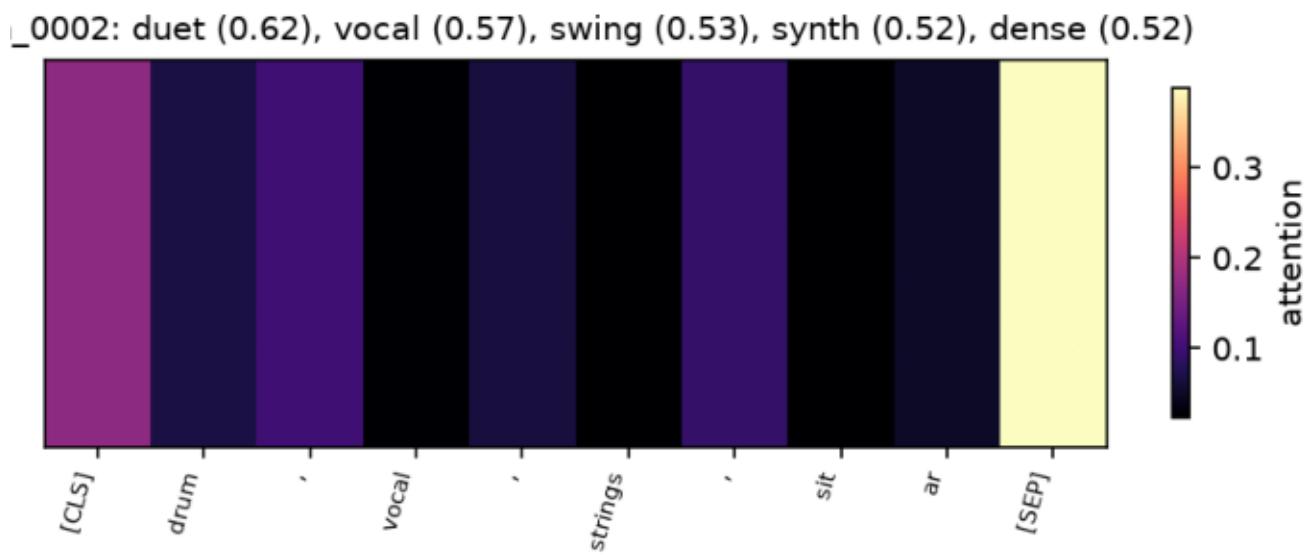
Plot types written: ['ablation', 'bert_attention', 'case_study', 'confusion_topk',
'f1_vs_epoch', 'graph_coherence', 'per_tag_prf', 'retrieval', 'seed_summary', 'tsne_genre',
'tsne_mood']

Built 2026-09-05 by report/build_report.py

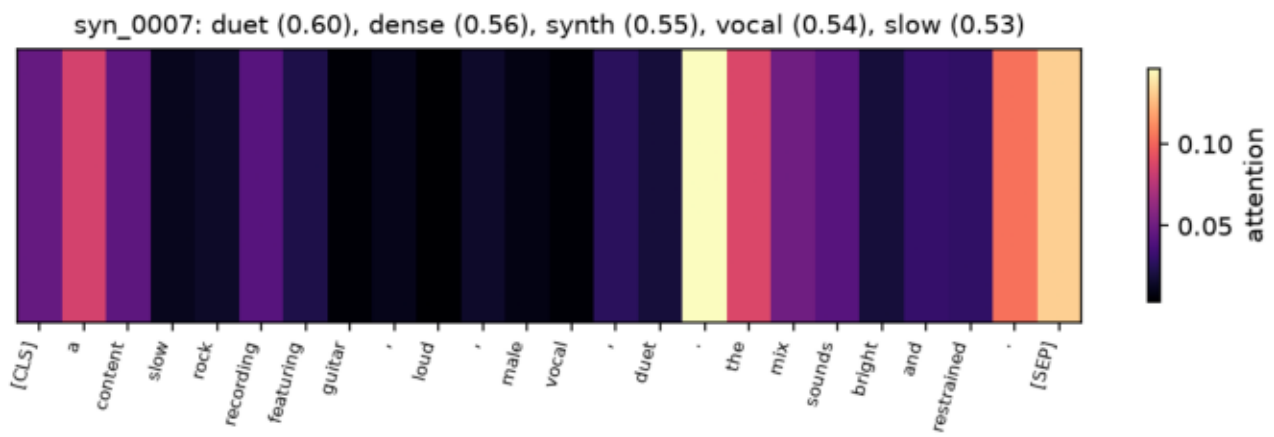
ablation



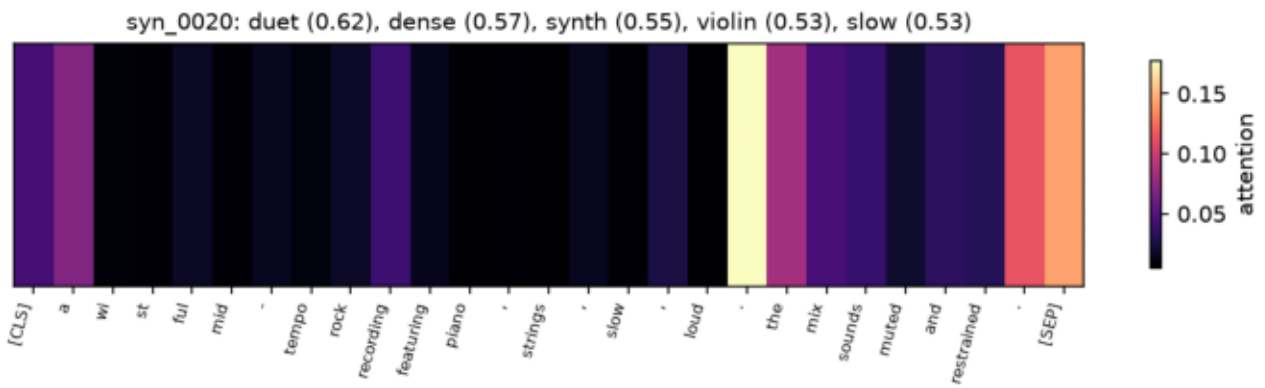
bert attention 00



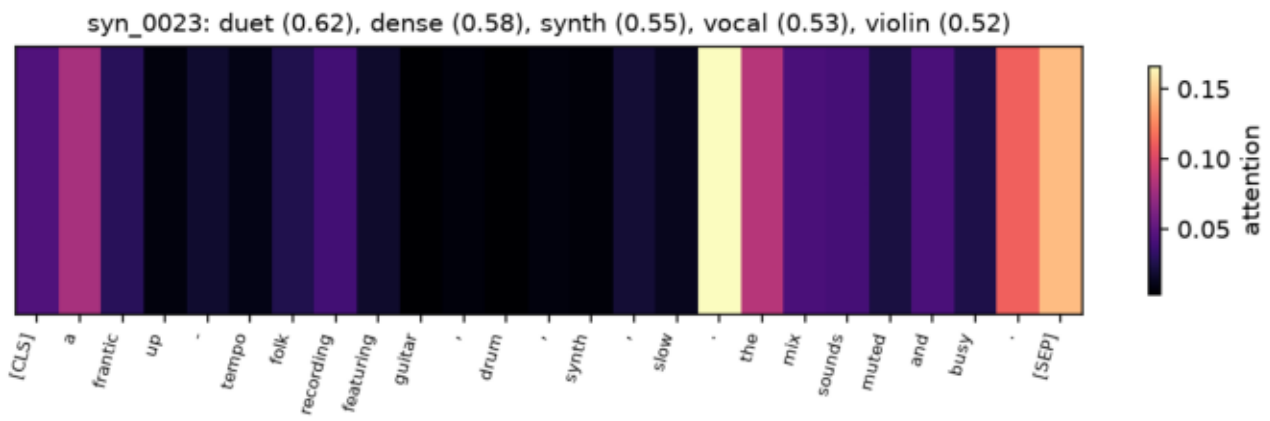
bert attention 01



bert attention 02

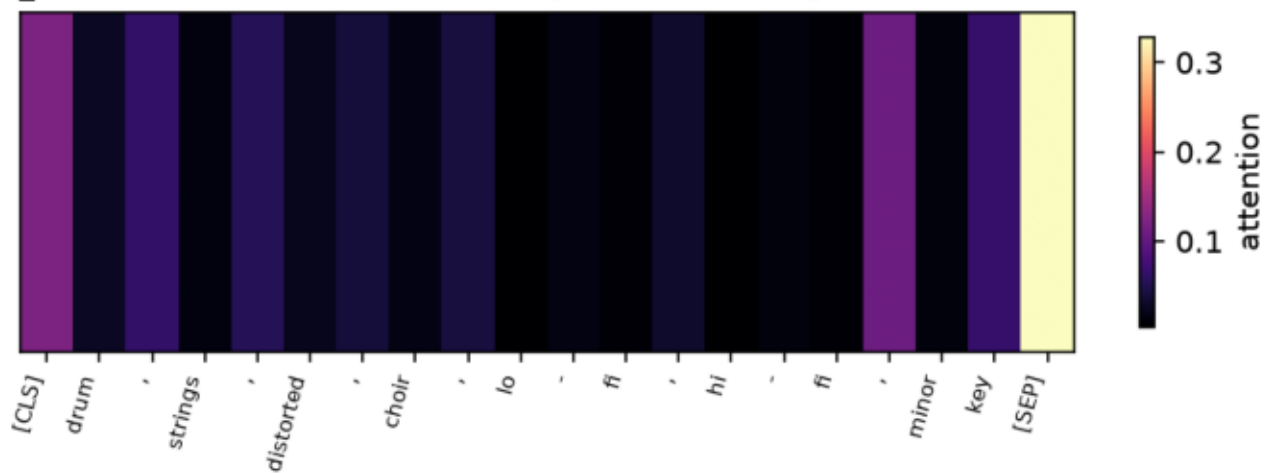


bert attention 03

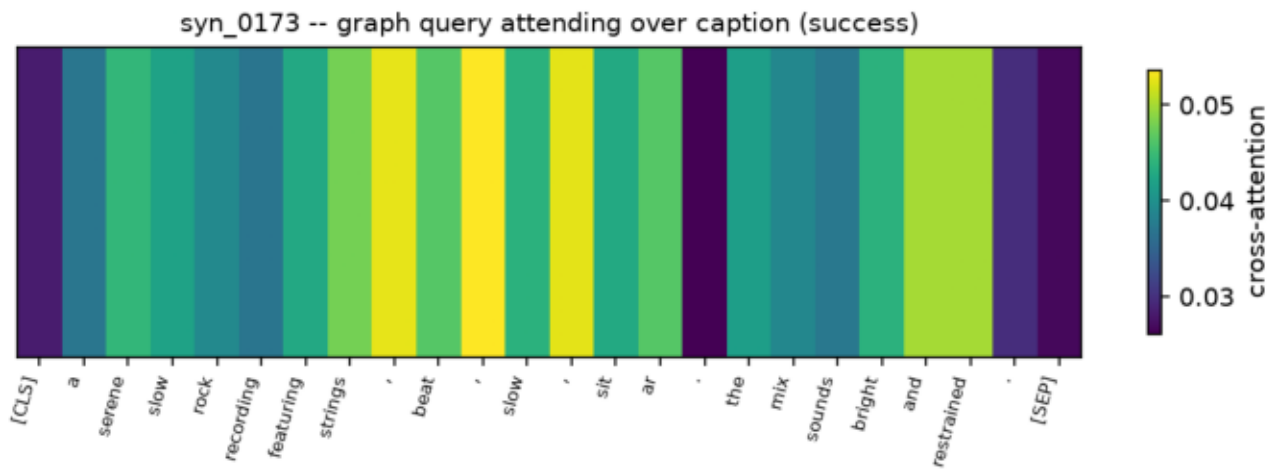


bert attention 04

syn_0040: duet (0.61), dense (0.54), synth (0.53), swing (0.52), vocal (0.51)

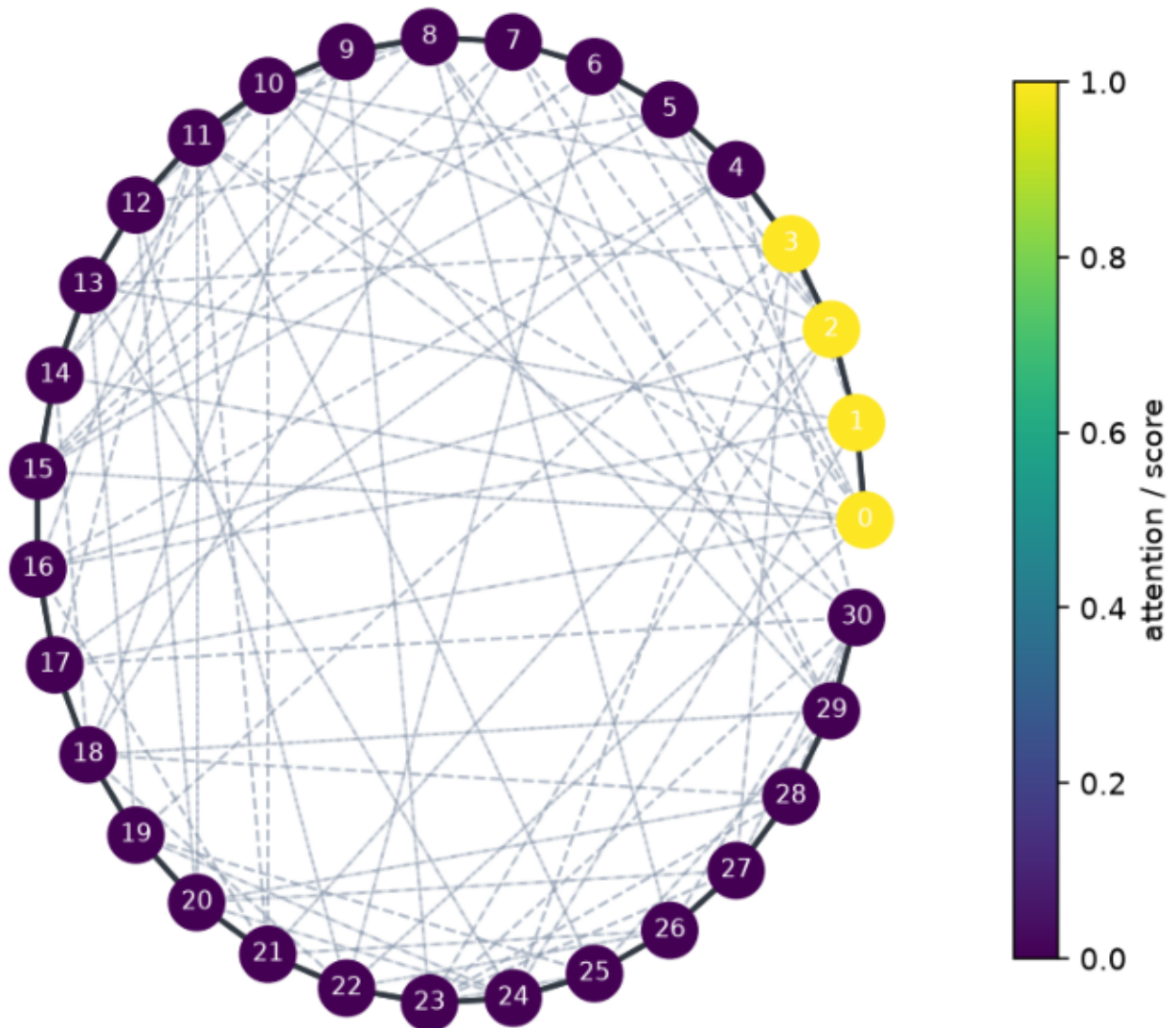


case study 0 cross attention

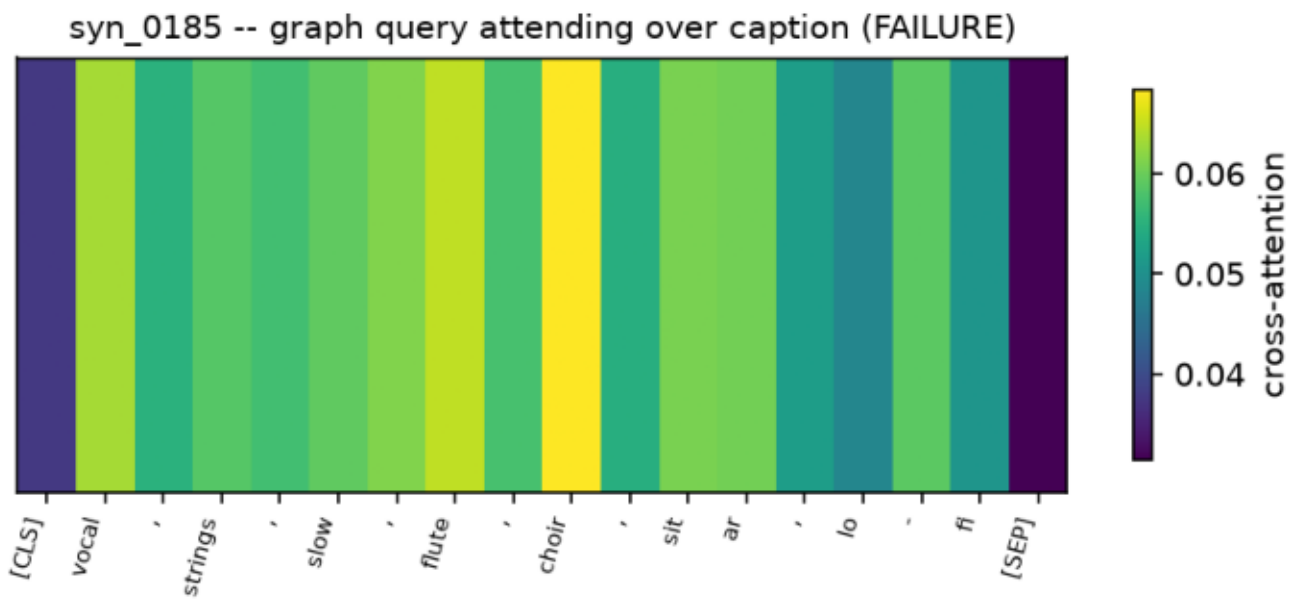


case study 0 graph

syn_0173 -- GATv2 attention per segment
31 segments, 107 undirected edges

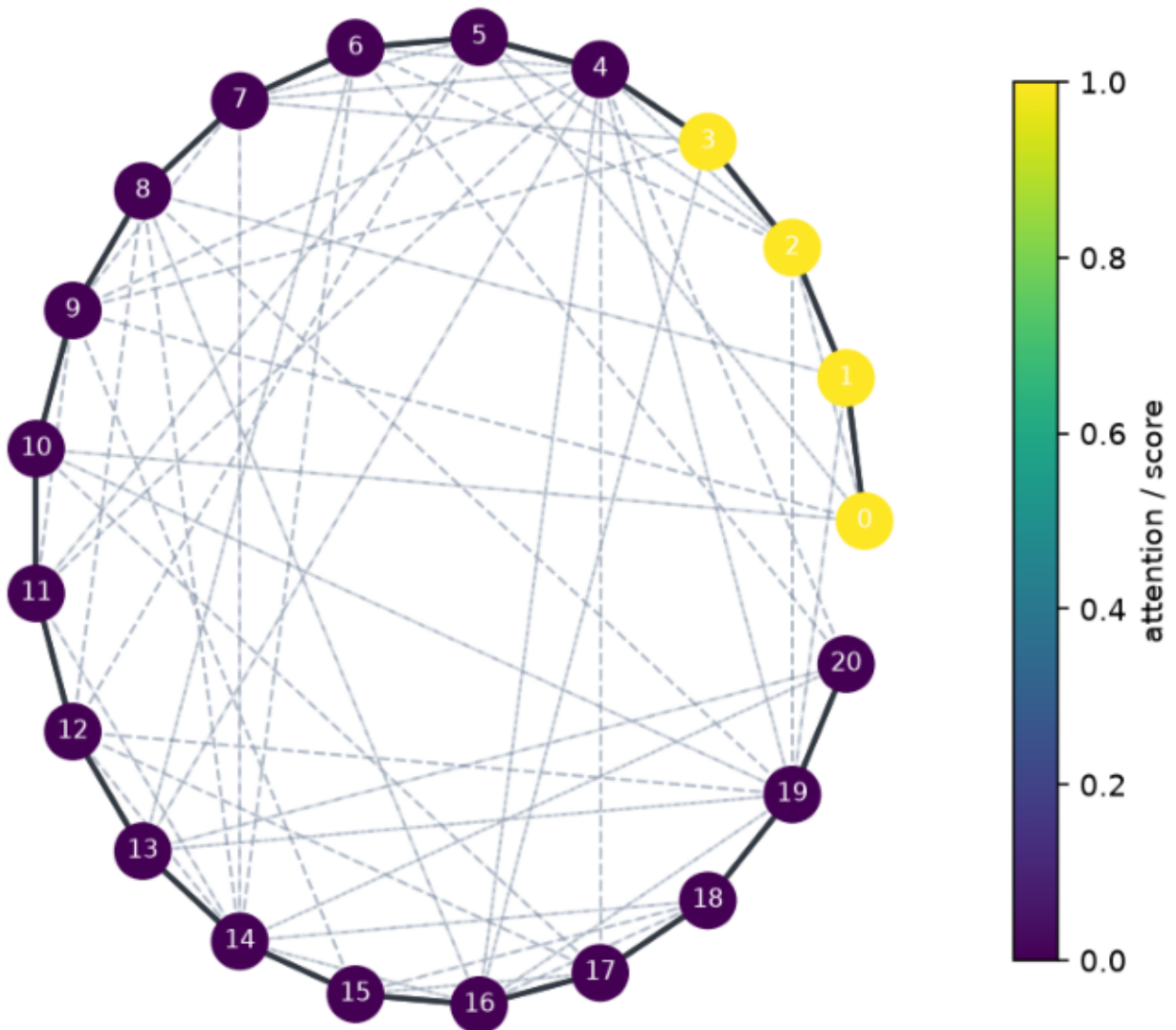


case study 1 cross attention

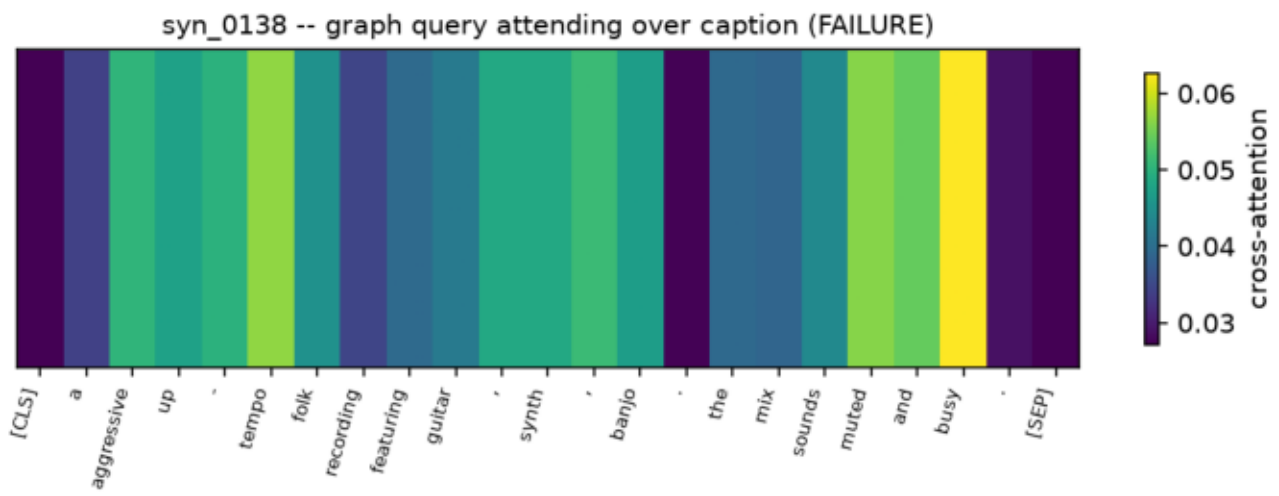


case study 1 graph

syn_0185 -- GATv2 attention per segment
21 segments, 71 undirected edges

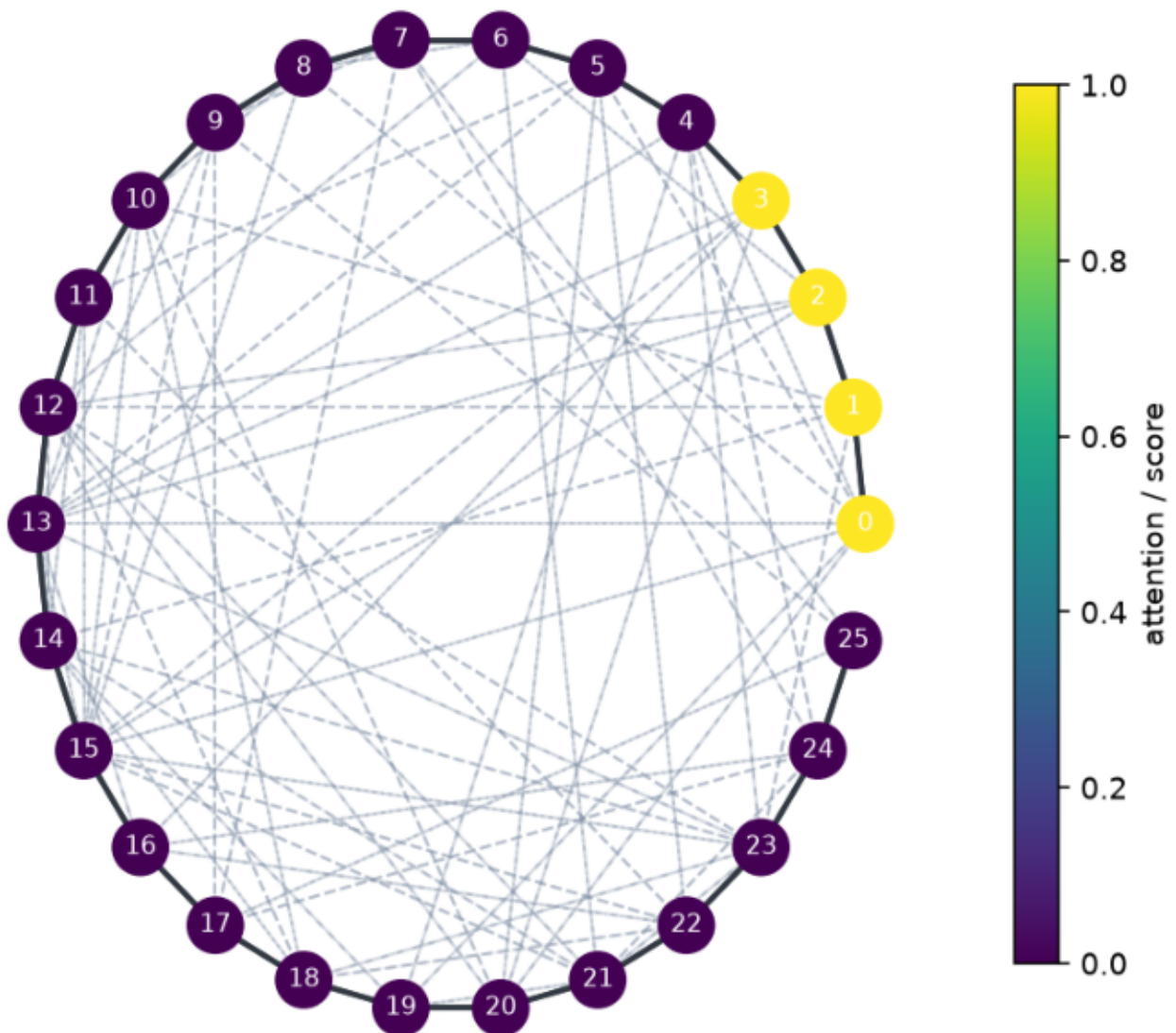


case study 2 cross attention



case study 2 graph

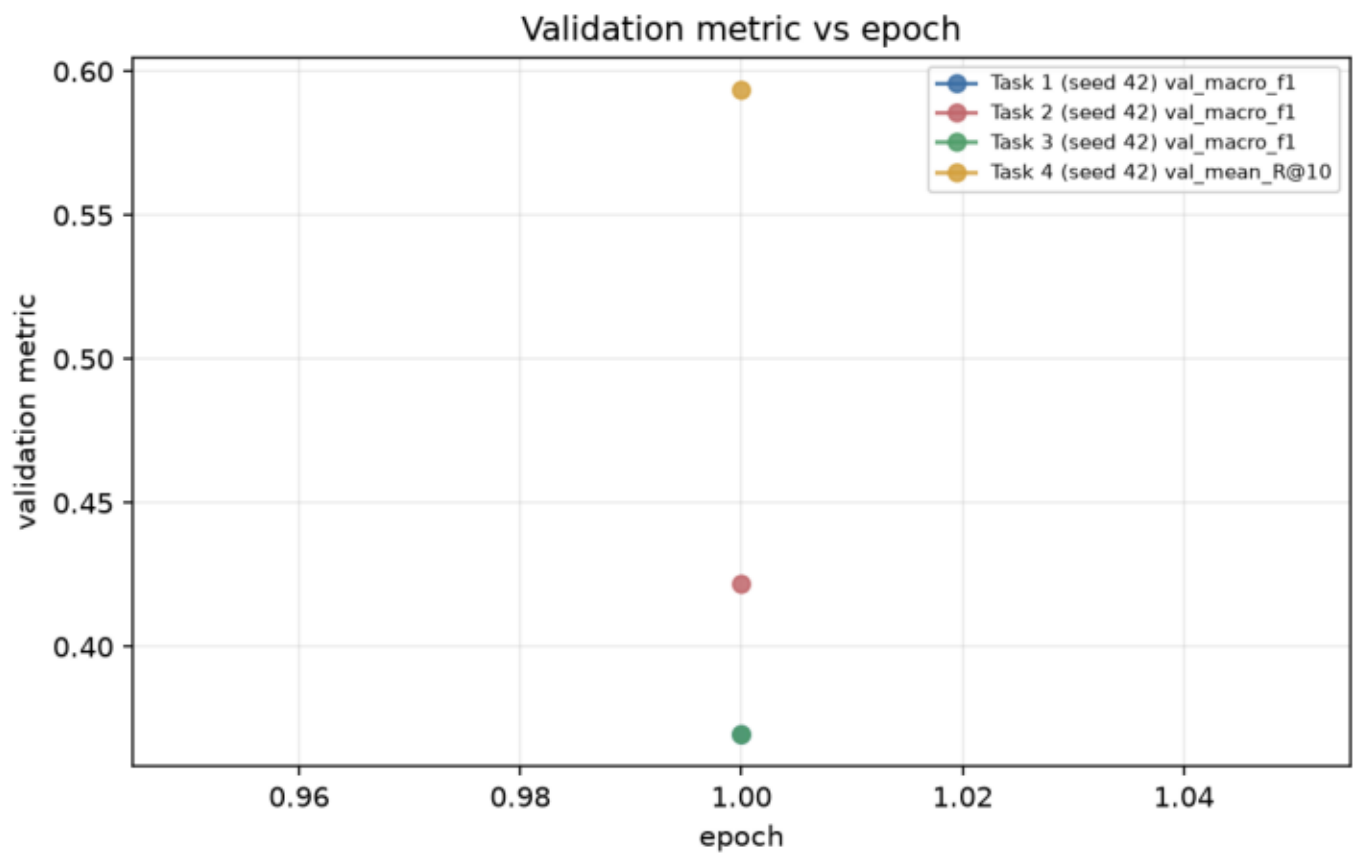
syn_0138 -- GATv2 attention per segment
26 segments, 96 undirected edges



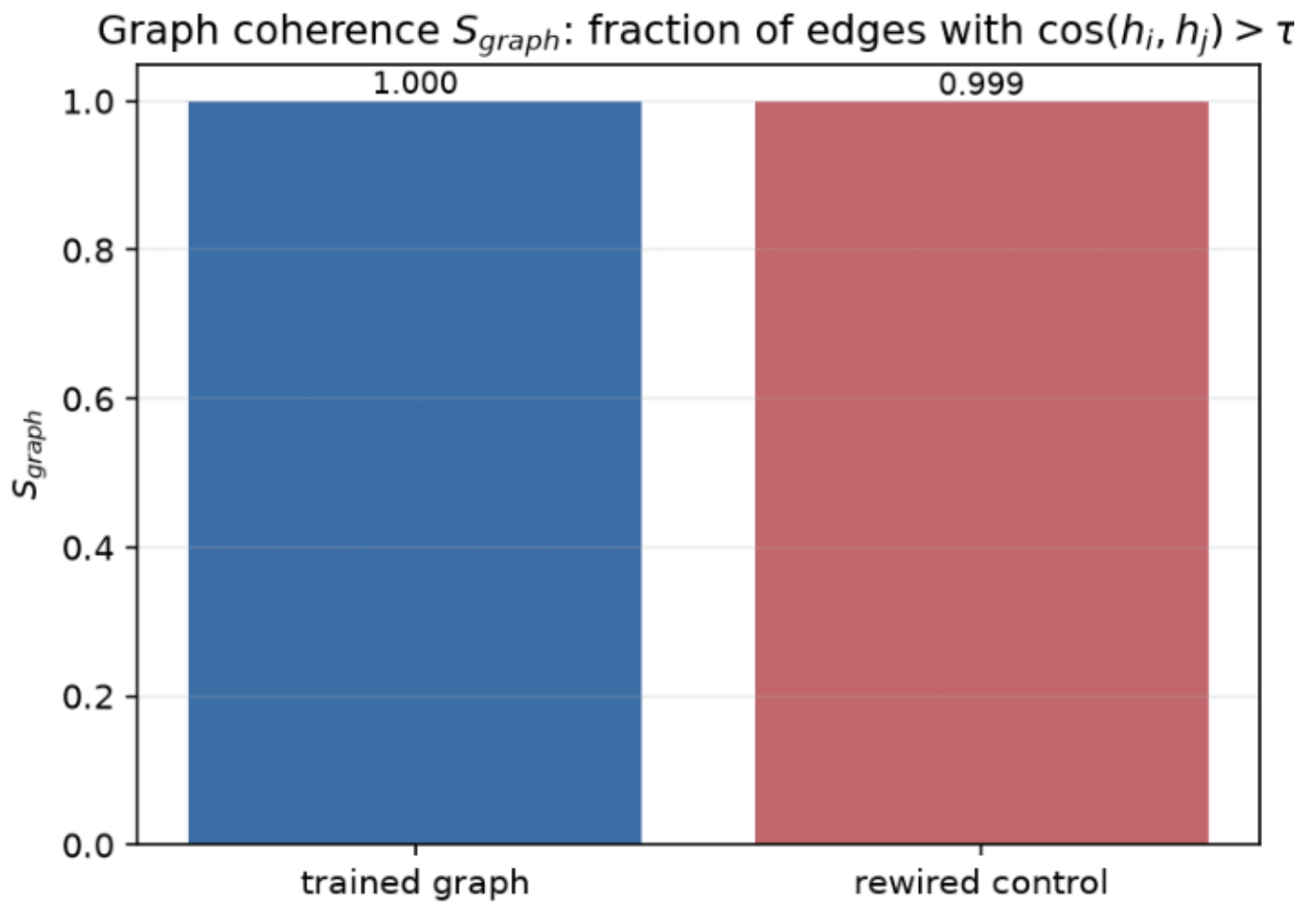
confusion topk



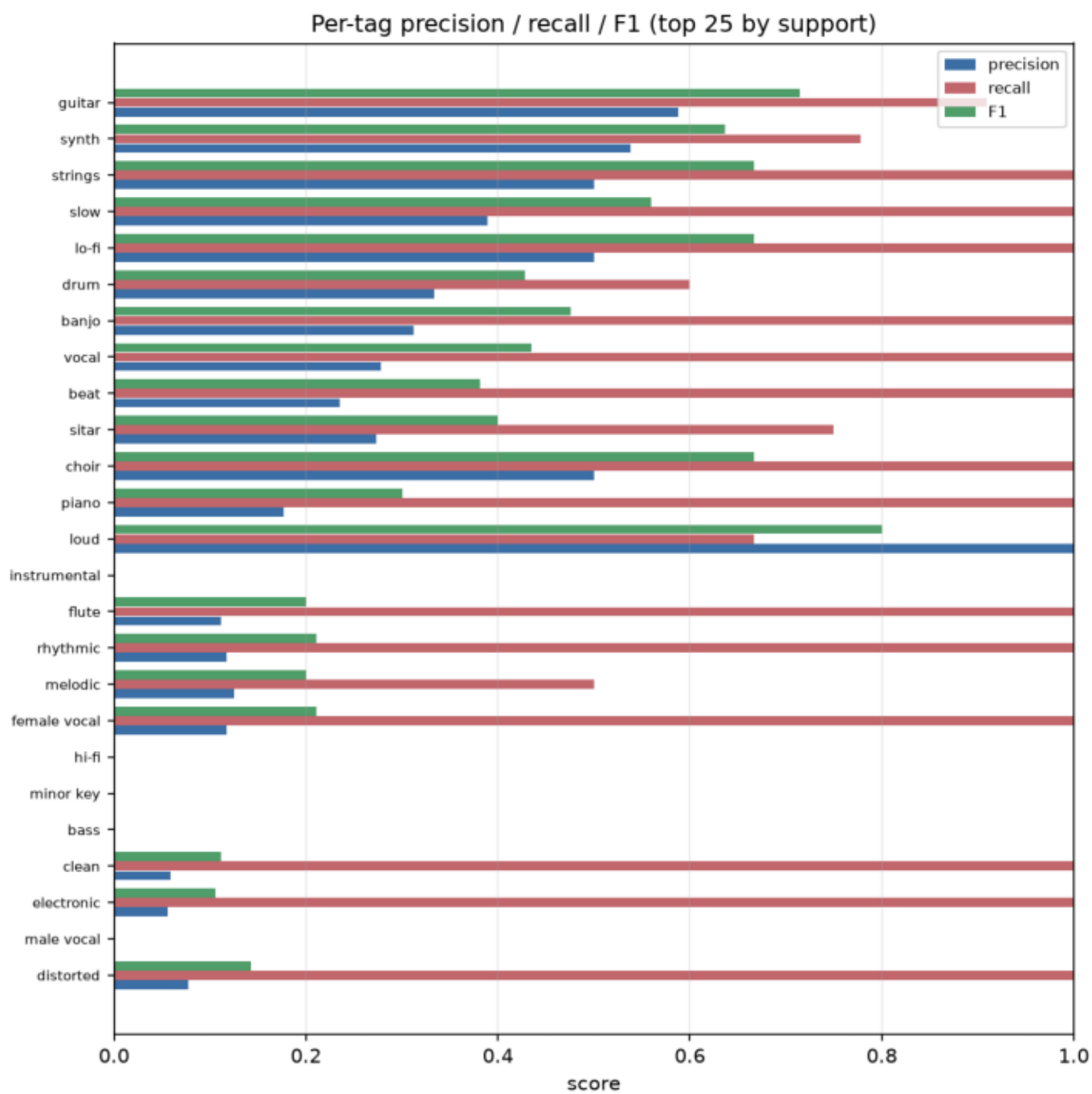
f1 vs epoch



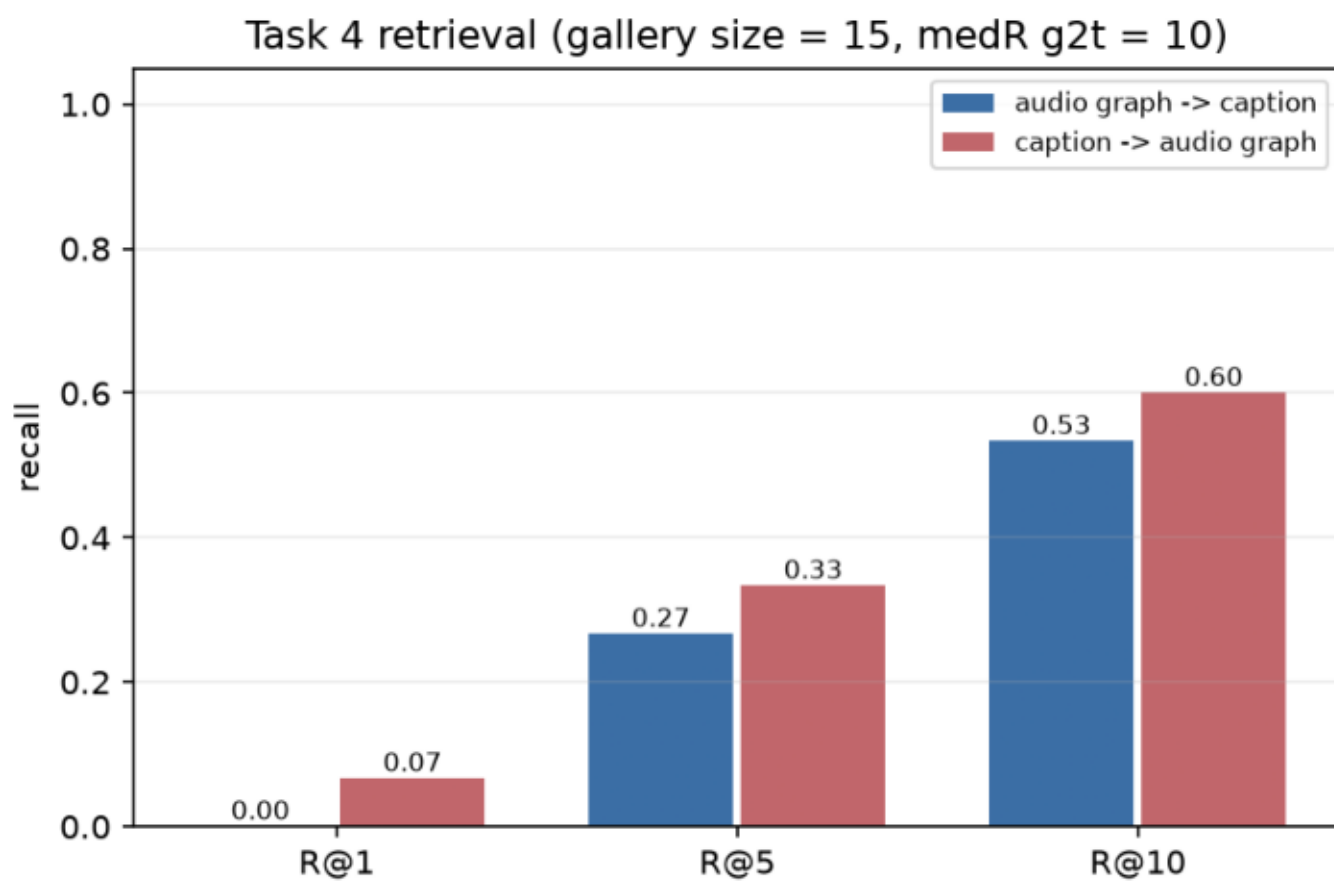
graph coherence



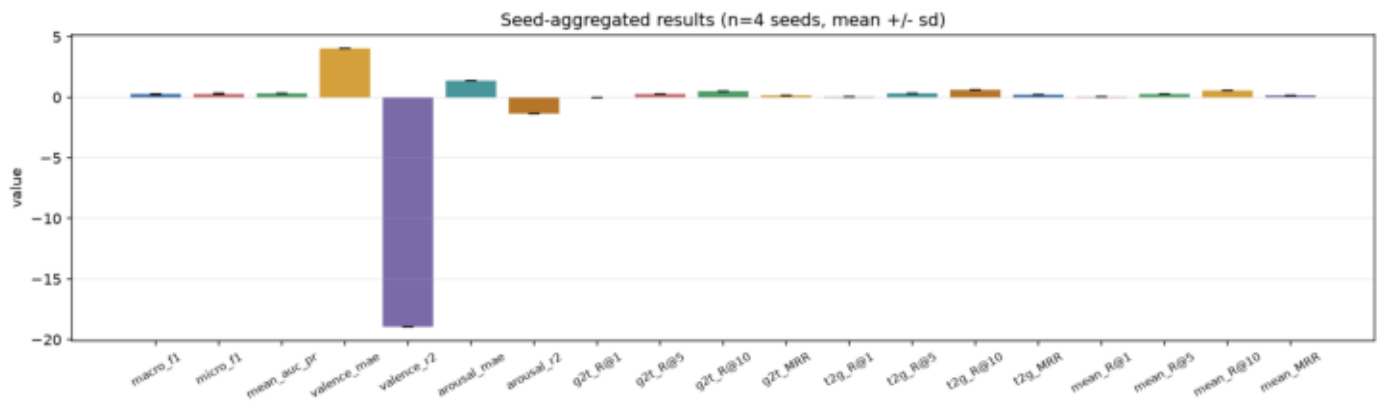
per tag prf



retrieval

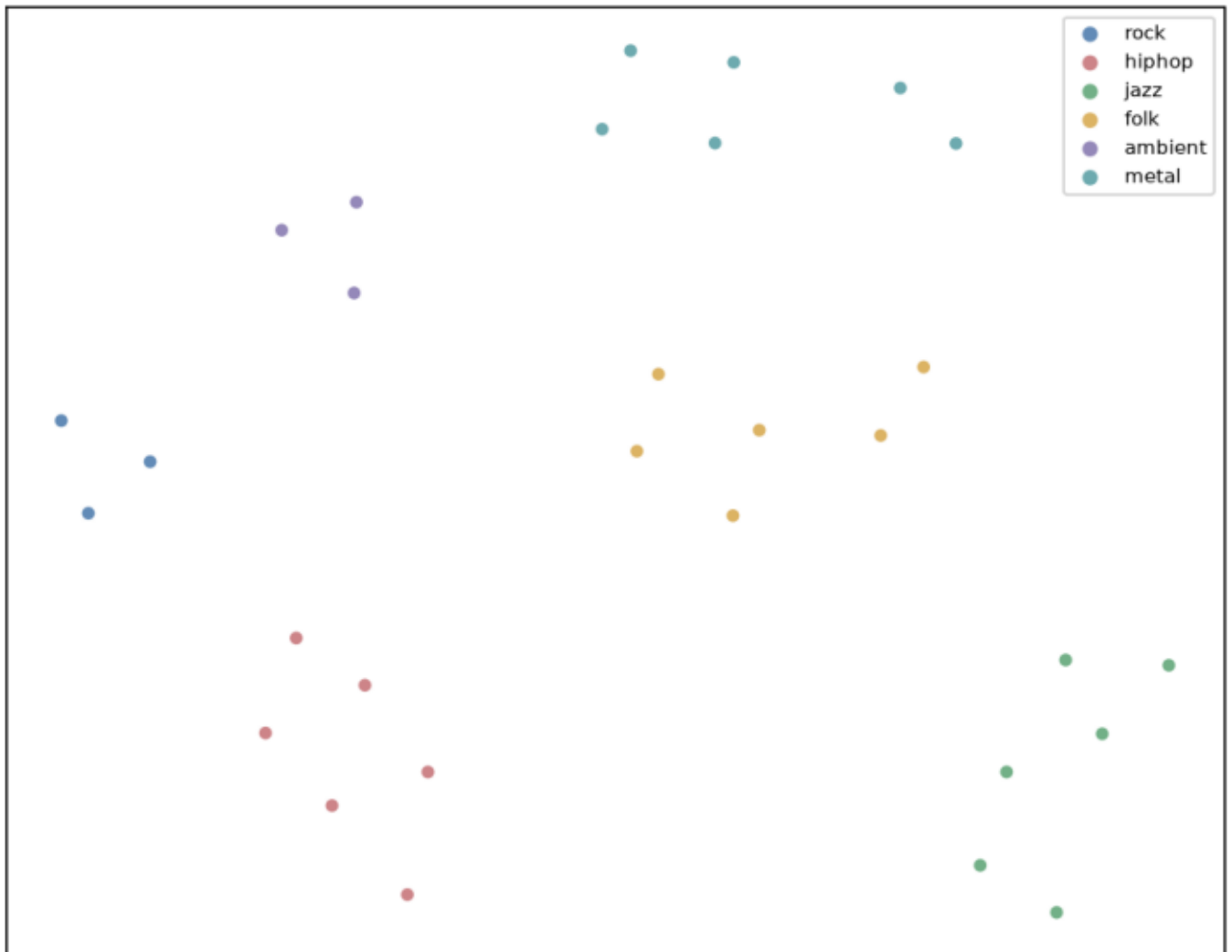


seed summary



tsne genre

Fused representation coloured by genre
perplexity=9 k-NN probe=0.800 silhouette=0.545



tsne mood

Fused representation coloured by mood quadrant
perplexity=5 k-NN probe=0.667 silhouette=0.277

